🏆 Infosys SE Elite Crack Vault — 09 Master Infosys Puzzle Solving Exam Vault Elite

🎯 Infosys Systems Engineer (SE) | 📊 35 Questions | 📅 5-6 Nov 2026 Drive | ⚡ 100% Pure SE Verified

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Problem 1 • Triangle Product and Top Addition Pattern (

🔥 MUST-SOLVE

`ID: Q1` `Source: Infosys PYQ` `Level: Infosys SE Standard`

 Figure

Question:

What will replace the question mark (?) in the third triangle?

Study the relationship between the three outer numbers (Top, Bottom-Left, Bottom-Right) and the central number in the first two triangles:
- Triangle 1: Top = 7, Bottom-Left = 6, Bottom-Right = 2, Center = 19
- Triangle 2: Top = 1, Bottom-Left = 8, Bottom-Right = 4, Center = 33
- Triangle 3: Top = 5, Bottom-Left = 2, Bottom-Right = ?, Center = 23

- A. 8
- B. 9
- C. 6
- D. 5

✓ **Verified Answer:** Answer specified in derivation below

Problem 2 • Concentric Hexagonal Circle Grid End-Average Pattern (

MUST-SOLVE

ID: Q2 | Source: Infosys PYQ | Level: Infosys SE Standard

 Figure

Question:
What number comes inside the circle marked with a question mark (?) in the concentric grid?

Grid layout across 5 horizontal levels:
- Level 1: 4, 3, 2
- Level 2: 5, 3, 5, 1, 1
- Level 3: 6, 1, 2, 8, 3, 3, 1
- Level 4: 7, 2, 8, 4, 3
- Level 5: 9, ?, 3

- A. 2
- B. 6
- C. 5
- D. 9

✓ **Verified Answer:** Answer specified in derivation below

Problem 3 • Octagonal Ring Reverse Alphabet Complement Pattern (

MUST-SOLVE

ID: Q3 | Source: Infosys PYQ | Level: Infosys SE Standard

 Figure

Question:
Which letter replaces the question mark (?) in the octagonal concentric ring?

Outer ring letters: D, G, Q, ?, Y, R, X, N
Inner ring letters: W, T, J, L, B, I, C, M

- A. O
- B. L
- C. Y
- D. P

✓ **Verified Answer:** Answer specified in derivation below

Problem 4 • Sequential Square Two-Digit Concatenation Pattern (

MUST-SOLVE

ID: Q4 Source: Infosys PYQ Level: Infosys SE Standard

 Figure

Question:
Which number replaces the question mark (?) in the paired octagon series?

Given pairs:
- Pair 1: (3, 6)
- Pair 2: (4, 9)
- Pair 3: (6, 4)
- Pair 4: (8, ?)

- A. 1
- B. 2
- C. 5
- D. 9

✓ **Verified Answer:** Answer specified in derivation below

Problem 5 • Recursive Progression Tree Pattern (2x + 3) (

MUST-SOLVE

ID: Q5 Source: Infosys PYQ Level: Infosys SE Standard

 Figure

Question:
Which number replaces the question mark (?) in the radial tree structure?

Nodes in clockwise sequence:
- Top-Left: 1
- Top-Center: 5
- Top-Right: 13
- Center Hub: 29
- Bottom-Left: 61
- Bottom-Center: 125
- Bottom-Right: ?

- A. 452
- B. 250
- C. 235
- D. 253

✓ **Verified Answer:** Answer specified in derivation below

Problem 6 • Two-Column Ladder Incremental Row Sum Pattern (

MUST-SOLVE

ID: Q6 | Source: Infosys PYQ | Level: Infosys SE Standard

 Figure

Question:
Which number replaces the question mark (?) in the 2-column segmented ladder?

Header: 10
- Row 1: [8 | 13]
- Row 2: [16 | 6]
- Row 3: [4 | 19]
- Row 4: [22 | ?]

- A. 24
- B. 4
- C. 2
- D. 8

✓ **Verified Answer:** Answer specified in derivation below

Problem 7 • Five-Pointed Star Upper Minus Lower Balance Pattern (

MUST-SOLVE

ID: Q7 | Source: Infosys PYQ | Level: Infosys SE Standard

 Figure

Question:
Which number replaces the question mark (?) in the center of the third 5-pointed star?

- Star 1: Top = 2, Left = 4, Right = 7, Bottom-Left = 3, Bottom-Right = 1, Center = 9
- Star 2: Top = 3, Left = 3, Right = 5, Bottom-Left = 4, Bottom-Right = 2, Center = 5
- Star 3: Top = 9, Left = 6, Right = 2, Bottom-Left = 4, Bottom-Right = 3, Center = ?

- A. 8
- B. 10
- C. 12
- D. 16

✓ **Verified Answer:** Answer specified in derivation below

Problem 8 • Reverse Alphabetical Positional Sum Box Pattern (MUST-SOLVE

ID: Q8 Source: Infosys PYQ Level: Infosys SE Standard

 Figure

Question:
Which letter replaces the question mark (?) in the 5×3 grid of boxes?

Rows:
- Row 1: [3 | P | 8]
- Row 2: [9 | G | 11]
- Row 3: [2 | U | 4]
- Row 4: [3 | W | 1]
- Row 5: [7 | ? | 18]

- A. B
- B. Y
- C. Z
- D. C

✓ **Verified Answer:** Answer specified in derivation below

Problem 9 • 4×3 Quadratic Sum Matrix Pattern ($A^2 + B^2 - 2 = C$) (MUST-SOLVE

ID: Q9 Source: Infosys PYQ Level: Infosys SE Standard

 Figure

Question:
Study the 4×3 matrix. Each row follows the same rule relating the first two entries to the third entry. Find the missing number (?) that makes the fourth row obey the same rule.

Matrix entries:
- Row 1: 2, 3, 11
- Row 2: 4, 1, 15
- Row 3: 7, 3, 56
- Row 4: ?, 8, 87

- A. 3
- B. 4
- C. 5
- D. 8

✓ **Verified Answer:** Answer specified in derivation below

Problem 10 • 3×3 Letter-Number Product Matrix Pattern (

MUST-SOLVE

ID: Q10 Source: Infosys PYQ Level: Infosys SE Standard

 Figure

Question:
Study the 3×3 matrix. Each column follows the same rule relating the top and bottom numerical entries to the middle letter entry. Find the missing letter (?) that makes the middle column obey the same rule.

Matrix entries:
- Row 1: 2 | 3 | 7
- Row 2: H | ? | U
- Row 3: 4 | 6 | 3

- A. P
- B. Q
- C. S
- D. R

✓ **Verified Answer:** Answer specified in derivation below

Problem 11 • Divided Circles Sum of Squares Pattern (

MUST-SOLVE

ID: Q11 Source: Infosys PYQ Level: Infosys SE Standard

 Figure

Question:
Find the missing term (?) in the third figure. Assume the third figure works in the same way as the other two.

Divided Circles:
- Circle 1: Top-Left = 2, Top-Right = 4, Bottom = 20
- Circle 2: Top-Left = 3, Top-Right = 9, Bottom = 90
- Circle 3: Top-Left = 1, Top-Right = 5, Bottom = ?

- A. 50
- B. 110
- C. 61
- D. 26

✓ **Verified Answer:** Answer specified in derivation below

Problem 12 • Dual-Variable Block Proportionality Pattern (

MUST-SOLVE

ID: Q12 Source: Infosys PYQ Level: Infosys SE Standard

 Figure

Question:
Find the missing terms (A and B) in the third figure. Assume the third figure works in the same way as the other two.

Block figures:
- Block 1: Top-Left = 14, Top-Right = 11, Bottom = 22
- Block 2: Top-Left = 12, Top-Right = 10, Bottom = 20
- Block 3: Top-Left = 20, Top-Right = A, Bottom = B

- A. A=9, B=17
- B. A=14, B=22
- C. A=14, B=28
- D. A=18, B=9

✓ **Verified Answer:** Answer specified in derivation below

Problem 13 • 4×4 Column Linear Product Matrix Pattern (

CORE SE

ID: Q13 Source: Infosys PYQ Level: Infosys SE Standard

 Figure

Question:
Study the 4×4 matrix. Each column follows the same rule relating the entries. Find the missing number (?) in the second row of the third column.

Matrix entries:
- Row 1: 11 | 4 | 21 | 39
- Row 2: 8 | 1 | ? | 2
- Row 3: 5 | 7 | 3 | 2
- Row 4: 63 | 29 | 72 | 80

- A. 7
- B. 9
- C. 5
- D. 8

✓ **Verified Answer:** Answer specified in derivation below

Problem 14 • Dual-Set Exponential Square Grid Pattern (

CORE SE

ID: Q14 Source: Infosys PYQ Level: Infosys SE Standard

 Figure

Question:
Determine the missing number (?) in the second set. Assume the second set also works in the same way as the first set.

Set 1 (3×3):
- Row 1: 3 | 2 | 8
- Row 2: 1 | 59 | 1
- Row 3: 5 | 14 | 25

Set 2 (3×3):
- Row 1: 4 | 11 | 16
- Row 2: 7 | ? | 49
- Row 3: 2 | 6 | 4

- A. 89
- B. 79
- C. 99
- D. 109

✓ **Verified Answer:** Answer specified in derivation below

Problem 15 • Segmented Four-Square Constant Sum (22) Pattern (

CORE SE

ID: Q15 Source: Infosys PYQ Level: Infosys SE Standard

 Figure

Question:
Which number replaces the question mark (?) in the segmented four-square ring?

- Top-Left Square: Top = 2, Left = 7, Right = 4, Bottom = 9
- Top-Right Square: Top = 11, Left = 3, Right = 2, Bottom = 6
- Bottom-Left Square: Top = 5, Left = 5, Right = 6, Bottom = 6
- Bottom-Right Square: Top = 1, Left = 8, Right = 4, Bottom = ?

- A. 7
- B. 8
- C. 5
- D. 9

✓ **Verified Answer:** Answer specified in derivation below

Problem 16 • T-Shaped Block Upper Mean Average Pattern (

CORE SE

ID: Q16 Source: Infosys PYQ Level: Infosys SE Standard

Figure

Question:
Which number replaces the question mark (?) in the fourth T-shaped block?

- Block 1: Top-Left = 7, Top-Right = 3, Bottom = 5
- Block 2: Top-Left = 8, Top-Right = 6, Bottom = 7
- Block 3: Top-Left = 2, Top-Right = 10, Bottom = 6
- Block 4: Top-Left = 4, Top-Right = 2, Bottom = ?

- A. 2
- B. 3
- C. 4
- D. 7

✓ **Verified Answer:** Answer specified in derivation below

Problem 17 • Hexagonal Letter Positional Addition Pattern (

CORE SE

ID: Q17 Source: Infosys PYQ Level: Infosys SE Standard

Figure

Question:
Which letter replaces the question mark (?) in the bottom row of hexagons?

Hexagon grid:
- Row 1: [E] [M] [H]
- Row 2: [N] [O] [A]
- Row 3: [I] [?] [D]

- A. M
- B. N
- C. O
- D. P

✓ **Verified Answer:** Answer specified in derivation below

Problem 18 • Team Selection Constraint Satisfaction Puzzle (CORE SE

ID: Q18 Source: Infosys PYQ Level: Infosys SE Standard

 Figure

Question:
A team of four players are to be selected from among four girls A, B, C, D and four boys P, Q, R, S such that there would be at least two girls in the team and the team selection should meet the following conditions:
i. B cannot go with R
ii. D and P must go together
iii. S cannot be put up with A
iv. R and C must be together

Which of the following is an acceptable team?

- A. QBSC
- B. DAPQ
- C. RCQS
- D. BPRA

✓ **Verified Answer:** Answer specified in derivation below

Problem 19 • Balbir Butcher Shop Dog Theft (Truth and Lie Logic) (CORE SE

ID: Q19 Source: Infosys PYQ Level: Infosys SE Standard

 Figure

Question:
While Balbir had his back turned, a dog ran into his butcher shop, snatched a piece of meat off the counter and ran off. Balbir was mad when he realised what had happened. He asked three other shopkeepers, who had seen the dog, to describe it. The shopkeepers really didn't want to help Balbir. So each of them made a statement which contained exactly one truth and one lie:

- Shopkeeper 1 said: 'The dog had black hair and a long tail.'
- Shopkeeper 2 said: 'The dog had a short tail and wore a collar.'
- Shopkeeper 3 said: 'The dog had white hair and no collar.'

Based on the above statements, which of the following could be a correct description?

- A. The dog had black hair, long tail and a collar.
- B. The dog had white hair, short tail and no collar.
- C. The dog had black hair, long tail and no collar.
- D. The dog had white hair, long tail and a collar.

✓ **Verified Answer:** Answer specified in derivation below

Problem 20 • The 3 Jug Water Riddle (12L, 8L, 5L State Transition) (

CORE SE

ID: Q20 Source: Infosys PYQ Level: Infosys SE Standard

 Figure

Question:
A milkman carries a full 12-liter container of milk. He needs to deliver exactly 6 liters to a customer who only has an 8-liter container and a 5-liter container. Neither container has any measurement markings. Can he measure exactly 6 liters without spilling or wasting any milk, and what is the minimum sequence of pours?

- A. Impossible without a 6-liter measuring vessel
- B. Possible in 7 pours: [12, 0, 0] -> [6, 6, 0]
- C. Possible in 9 pours: [12, 0, 0] -> [6, 1, 5]
- D. Possible in 5 pours

✓ **Verified Answer:** Answer specified in derivation below

Problem 21 • The 8 Balls Balance Scale Puzzle (

CORE SE

ID: Q21 Source: Infosys PYQ Level: Infosys SE Standard

 Figure

Question:
You have a two-pan balance weighing scale and 8 identical-looking balls. Except for one defective ball which is slightly heavier than the others, all other 7 balls have identical weights. What is the minimum number of weighings required on the balance scale to guarantee finding the defective ball in all cases?

- A. 1
- B. 2
- C. 3
- D. 4

✓ **Verified Answer:** Answer specified in derivation below

Problem 22 • Infosys PYQ Matrix Deduction Problem 22 (

CORE SE

ID: Q22 | Source: Infosys PYQ | Level: Infosys SE Standard

 Figure

Question:

1. Find the missing number (?) from the given matrix: **Native Grid Table:**

18	16	34	16	20	36

Native Grid Table:

18	16	34	16	20	36

[Grid: 18 | 16 | 34 | 16 | 20 | 36]

A 6

B 15

C 30

D 18

✓ **Verified Answer:** `D` **Official Solution Proof:** Step-by-step matrix deduction: 18 16 34 Conclusion confirms answer D.

Problem 23 • Infosys PYQ Matrix Deduction Problem 23 (

CORE SE

ID: Q23 | Source: Infosys PYQ | Level: Infosys SE Standard

 Figure

Question:

2. Find the missing number (?) from the given matrix: **Native Grid Table:**

7	8	15	21	12	33

Native Grid Table:

7	8	15	21	12	33

[Grid: 7 | 8 | 15 | 21 | 12 | 33]

A 34

B 31

C 19

D 43

✓ **Verified Answer:** `B` **Official Solution Proof:** Step-by-step matrix deduction: 7 8 15 Conclusion confirms answer B.

Problem 24 • Infosys PYQ Matrix Deduction Problem 24 (

CORE SE

ID: Q24 | Source: Infosys PYQ | Level: Infosys SE Standard

 Figure

Question:

3. Find the missing number (?) from the given matrix: **Native Grid Table:**

16	4	20	13	17	30

Native Grid Table:

16	4	20	13	17	30

[Grid: 16 | 4 | 20 | 13 | 17 | 30]

A 32

B 26

C 36

D 18

✓ **Verified Answer:** `B` **Official Solution Proof:** Step-by-step matrix deduction: 16 4 20 Conclusion confirms answer B.

Problem 25 • Infosys PYQ Matrix Deduction Problem 25 (

CORE SE

ID: Q25 | Source: Infosys PYQ | Level: Infosys SE Standard

 Figure

Question:

4. Find the missing number (?) from the given matrix: **Native Grid Table:**

16	9	25	13	15	28

Native Grid Table:

16	9	25	13	15	28

[Grid: 16 | 9 | 25 | 13 | 15 | 28]

A 42

B 30

C 25

D 48

✓ **Verified Answer:** `B` **Official Solution Proof:** Step-by-step matrix deduction: 16 9 25 Conclusion confirms answer B.

Problem 26 • Infosys PYQ Matrix Deduction Problem 26 (

CORE SE

ID: Q26 | Source: Infosys PYQ | Level: Infosys SE Standard

 Figure

Question:

5. Find the missing number (?) from the given matrix: **Native Grid Table:**

8	15	23	18	5	23

Native Grid Table:

8	15	23	18	5	23

[Grid: 8 | 15 | 23 | 18 | 5 | 23]

A 37

B 22

C 47

D 45

✓ **Verified Answer:** `A` **Official Solution Proof:** Step-by-step matrix deduction: 8 15 23 Conclusion confirms answer A.

Problem 27 • Infosys PYQ Matrix Deduction Problem 27 (

CORE SE

ID: Q27 | Source: Infosys PYQ | Level: Infosys SE Standard

 Figure

Question:

6. Find the missing number (?) from the given matrix: **Native Grid Table:**

25	15	10	33	11	22

Native Grid Table:

25	15	10	33	11	22

[Grid: 25 | 15 | 10 | 33 | 11 | 22]

A 4

B 21

C 13

D 16

✓ **Verified Answer:** `D` **Official Solution Proof:** Step-by-step matrix deduction: 25 15 10 Conclusion confirms answer D.

Problem 28 • Infosys PYQ Matrix Deduction Problem 28 (

CORE SE

ID: Q28 | Source: Infosys PYQ | Level: Infosys SE Standard

 Figure

Question:

7. Find the missing number (?) from the given matrix: **Native Grid Table:**

11	4	7	34	9	25

Native Grid Table:

11	4	7	34	9	25

[Grid: 11 | 4 | 7 | 34 | 9 | 25]

A 2

B 9

C 18

D 6

✓ **Verified Answer:** `D` **Official Solution Proof:** Step-by-step matrix deduction: 11 4 7 Conclusion confirms answer D.

Problem 29 • Infosys PYQ Matrix Deduction Problem 29 (

CORE SE

ID: Q29 | Source: Infosys PYQ | Level: Infosys SE Standard

 Figure

Question:

8. Find the missing number (?) from the given matrix: **Native Grid Table:**

30	8	22	25	5	20

Native Grid Table:

30	8	22	25	5	20

[Grid: 30 | 8 | 22 | 25 | 5 | 20]

A 23

B 22

C 11

D 17

✓ **Verified Answer:** `D` **Official Solution Proof:** Step-by-step matrix deduction: 30 8 22 Conclusion confirms answer D.

Problem 30 • Infosys PYQ Matrix Deduction Problem 30 (

CORE SE

ID: Q30 | Source: Infosys PYQ | Level: Infosys SE Standard

 Figure

Question:

9. Find the missing number (?) from the given matrix: **Native Grid Table:**

23	9	14	31	9	22

Native Grid Table:

23	9	14	31	9	22

[Grid: 23 | 9 | 14 | 31 | 9 | 22]

A 18

B 21

C 30

D 15

✓ **Verified Answer:** `D` **Official Solution Proof:** Step-by-step matrix deduction: 23 9 14 Conclusion confirms answer D.

Problem 31 • Infosys PYQ Matrix Deduction Problem 31 (

CORE SE

ID: Q31 | Source: Infosys PYQ | Level: Infosys SE Standard

 Figure

Question:

10. Find the missing number (?) from the given matrix: **Native Grid Table:**

15	5	10	22	9	13

Native Grid Table:

15	5	10	22	9	13

[Grid: 15 | 5 | 10 | 22 | 9 | 13]

A 29

B 23

C 19

D 11

✓ **Verified Answer:** `D` **Official Solution Proof:** Step-by-step matrix deduction: 15 5 10 Conclusion confirms answer D.

Problem 32 • Infosys PYQ Matrix Deduction Problem 32 (

CORE SE

ID: Q32 | Source: Infosys PYQ | Level: Infosys SE Standard

 Figure

Question:

11. Find the missing number (?) from the given matrix: **Native Grid Table:**

7	5	35	4	3	12

Native Grid Table:

7	5	35	4	3	12

[Grid: 7 | 5 | 35 | 4 | 3 | 12]

A 66

B 74

C 56

D 52

✓ **Verified Answer:** `C` **Official Solution Proof:** Step-by-step matrix deduction: 7 5 35 Conclusion confirms answer C.

Problem 33 • Infosys PYQ Matrix Deduction Problem 33 (

CORE SE

ID: Q33 | Source: Infosys PYQ | Level: Infosys SE Standard

 Figure

Question:

12. Find the missing number (?) from the given matrix: **Native Grid Table:**

9	3	27	4	8	32

Native Grid Table:

9	3	27	4	8	32

[Grid: 9 | 3 | 27 | 4 | 8 | 32]

A 12

B 2

C 17

D 16

✓ **Verified Answer:** `A` **Official Solution Proof:** Step-by-step matrix deduction: 9 3 27 Conclusion confirms answer A.

Problem 34 • Infosys PYQ Matrix Deduction Problem 34 (

CORE SE

ID: Q34 | Source: Infosys PYQ | Level: Infosys SE Standard

 Figure

Question:

13. Find the missing number (?) from the given matrix: **Native Grid Table:**

9	4	36	9	8	72

Native Grid Table:

9	4	36	9	8	72

[Grid: 9 | 4 | 36 | 9 | 8 | 72]

A 38

B 50

C 27

D 32

✓ **Verified Answer:** `D` **Official Solution Proof:** Step-by-step matrix deduction: 9 4 36 Conclusion confirms answer D.

Problem 35 • Infosys PYQ Matrix Deduction Problem 35 (

CORE SE

ID: Q35 | Source: Infosys PYQ | Level: Infosys SE Standard

 Figure

Question:

14. Find the missing number (?) from the given matrix: **Native Grid Table:**

2	4	8	6	6	36

Native Grid Table:

2	4	8	6	6	36

[Grid: 2 | 4 | 8 | 6 | 6 | 36]

A 54

B 36

C 33

D 40

✓ **Verified Answer:** `B` **Official Solution Proof:** Step-by-step matrix deduction: 2 4 8 Conclusion confirms answer B.